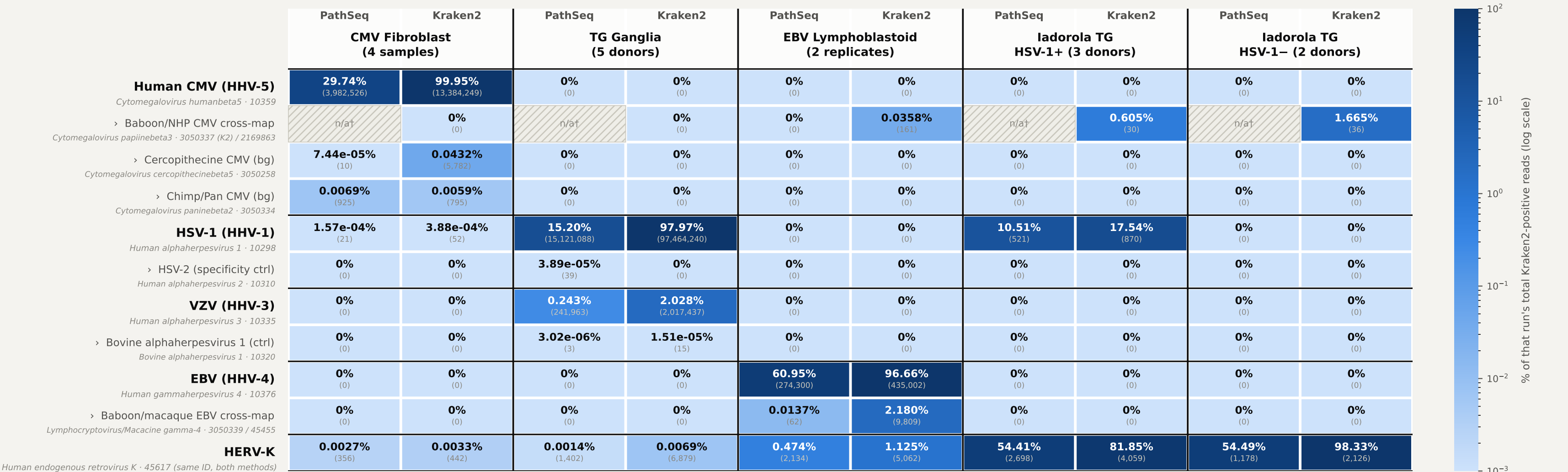


Kraken2 × PathSeq Concordance — Full High-Resolution

Rows grouped by taxonomic lineage — each main virus (bold) with every related sub-taxon indented directly beneath it. Now 5 cohorts: the original 3, plus the ladorola et al. TG cohort split into its own real Kraken2-confirmed HSV-1-positive and HSV-1-negative columns (not lumped as one cohort).



Normalization: for each column, % = (taxon's raw read count ÷ sum of Kraken2 raw reads across every taxon shown for that column) × 100, applied identically to both PathSeq and Kraken2. ladorola HSV+ = TG3+TG12+TG4 summed; HSV– = TG13+TG2 summed — real per-donor Tier-1 Kraken2 calls, PathSeq-confirmed concordant on every donor (see docs/pathseq_validation_results_2026-08-15.md). Full cross-validation complete for both new columns (2026-08-17) — every taxon in this figure checked against both tools. One real, non-zero finding: the CMV-proxy artifact (3050337) recurs here too, a 4th independent cohort showing it (after the original DRG/muscle samples, cmv_fibroblast's absence-as-control, and ebv_gm12878's trace background). Everything else in both new columns is a confirmed true-negative, not an unchecked assumption. QC note: three real output files for this one batch used three DIFFERENT sample column orders (see this script's module docstring) — every number here is header-verified against the actual file, not assumed from samplesheet order, after an initial mis-read briefly looked like a real Kraken2/PathSeq discordance. † PathSeq has no taxon distinct from 10359 representing Kraken2's specific proxy-species artifact— the Human CMV row's own value already answers the equivalent question in every cohort it applies to. Source: docs/pathseq_validation_results_2026-08-15.md, research/cmv_taxonomy_investigation.md.